

Programming Fundamentals

Spring 2026



Homework 2

Simple Problem Solving

Professor HG Locklear

Problem

- Tie Fighters routinely engage X-Wing Fighters in combat.
- Imperial Training Command has tasked us to create a Java program that models a simple simulation of this combat.

What Do We Know About How Combat is Resolved

- The Tie Fighter and the X-Wing Fighter **appear randomly** on the Battle Map (See Slide 3).
- The Tie Fighter **will not appear in the same location** as the X-Wing Fighter on the Battle Map.
- The distance between the Tie Fighter and the X-Wing Fighter is **calculated using 2D Euclidean distance**.
- The Fighters can engage each other **regardless of the distance** between them.
- Both the Tie Fighter and the X-Wing Fighter get to **shoot at each other three times** with the **Tie Fighter firing all three of its shots first** followed by the X-Wing Fighter.
- Each of the Fighters has a probability of hitting the other based the information shown below:

Range	1 - 3	4 - 6	7 - 8	> 8
Tie Fighter	70%	60%	50%	45%
X-Wing Fighter	85%	70%	60%	50%

Problem Graphic

	A	B	C	D
A	0,0	0,1	0,2	
B	1,0	1,1	1,2	1,3
C	2,0	2,1	2,2	2,3
D	3,0	3,1		3,3

Combat occurs on a **Battle Map**.
The map **consist of individual grids** and the map is either **4 or 8 grids in length**.

The **Battle Map is always square** having the same length and width.

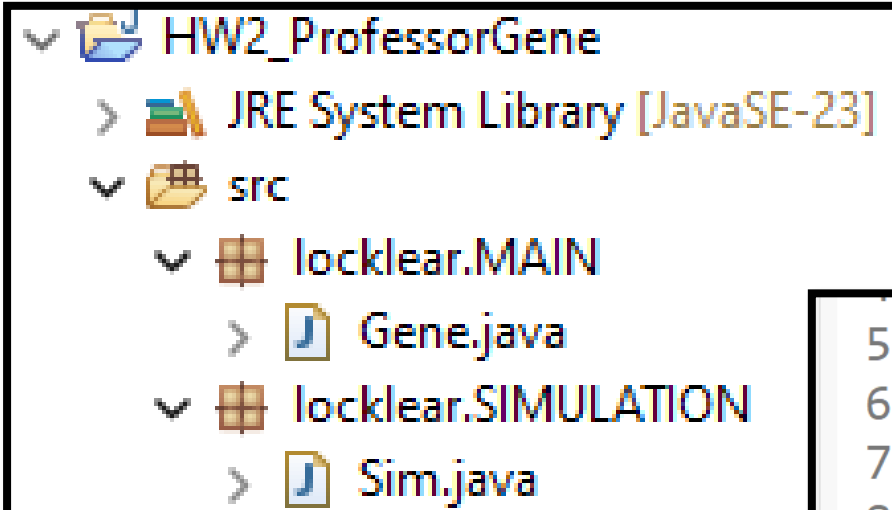
Battle Maps are **randomly created** for each combat resolution.

Battle Maps have **capital letter identifiers for each row and column** and **each grid has a X and Y coordinate** labelled as shown.

When the Battle Map is displayed, **the letters identifiers are displayed** but the X and Y coordinates are not, and each grid is displayed as [].

The location of the Tie Fighter on the Battle Map is shown as ***TF*** and the X-Wing Fighter is shown as ***XW***

Program Structure



Create your **program structure EXACTLY** as shown
replace my name with yours as appropriate.

Create your **Gene class EXACTLY** as shown

```
5 public class Gene {
6
7     public static String[][] bMap1 = Sim.battleMap(4);
8     public static String[][] bMap2 = Sim.battleMap(8);
9
10    public static void main(String[] args) {
11        System.out.println("***Engagement 1***");
12        Sim.AnalyzeFighterEngagement(bMap1);
13        System.out.println();
14        System.out.println("***Engagement 2***");
15        Sim.AnalyzeFighterEngagement(bMap2);
16    }
17
18 }
```

Tasks

- For us to create a solution to this problem, **we must accomplish two tasks:**
- **TASK 1**
- Create a static method **battleMap** that accepts the size of the Battle Map **returns a 2D String array** that **represents the Battle Map as defined on Slide 3.**
- **TASK 2**
- Create a static method **AnalyzeFighterEngagement** that accepts a Battle Map. The method simulates and **prints to the console** the results of combat between a single Tie Fighter and a single X-Wing Fighter based on the **information shown on Slide 2** and in the **format shown on Slides 6 and 7.**

Requirements and Guidelines

- You **may only use 1D and 2D** arrays for any data structures you need.
- The **battleMap** and **AnalyzeFighterEngagement** methods must be created in the **Sim** class.
- You may create **any number of helper methods**, but they **must all be created** in the **Sim** class.
- You **may not make any changes** to the **Gene** class and its **main method**.
- You are encouraged to make **extensive use of modularization** in your solution.

Expected Output

Problems @ Javadoc Declaration Console X

<terminated> Gene (5) [Java Application] C:\Users\msgth\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v20241024-170

Engagement 1

A

B

C

D

A [] [] [] []

B [] [] [] []

C [] [] [] *TF*

D [] *XW* [] []

Fighter	Fighter	Coords	Target	Target	Coords	Range	Shot	Accuracy	Hit-Miss
Tie		[2-3]	X-Wing		[3-1]	2.24	1	13	1
Tie		[2-3]	X-Wing		[3-1]	2.24	2	92	0
Tie		[2-3]	X-Wing		[3-1]	2.24	3	61	1
X-Wing		[3-1]	Tie		[2-3]	2.24	1	68	1
X-Wing		[3-1]	Tie		[2-3]	2.24	2	77	1
X-Wing		[3-1]	Tie		[2-3]	2.24	3	4	1

Accuracy is generated randomly for each shot and is representative of **how accurate the current shot was.**

Accuracy is **compared to the POH** (Slide 2) **for the target range** to determine if the **target has been hit or not.**

A **0** represents a **miss** and a **1** represents a **hit**

Expected Output

Engagement 2

	A	B	C	D	E	F	G	H
A	[]	[]	[]	[]	[]	[]	[]	[]
B	[]	[]	[]	[]	[]	[]	[]	[]
C	*TF*	[]	[]	[]	[]	[]	[]	[]
D	[]	[]	[]	[]	[]	[]	[]	[]
E	[]	[]	[]	[]	*XW*	[]	[]	[]
F	[]	[]	[]	[]	[]	[]	[]	[]
G	[]	[]	[]	[]	[]	[]	[]	[]
H	[]	[]	[]	[]	[]	[]	[]	[]

Fighter	Fighter	Coords	Target	Target	Coords	Range	Shot	Accuracy	Hit-Miss
Tie		[2-0]	X-Wing		[4-4]	4.47	1	83	0
Tie		[2-0]	X-Wing		[4-4]	4.47	2	0	1
Tie		[2-0]	X-Wing		[4-4]	4.47	3	16	1
X-Wing		[4-4]	Tie		[2-0]	4.47	1	81	0
X-Wing		[4-4]	Tie		[2-0]	4.47	2	5	1
X-Wing		[4-4]	Tie		[2-0]	4.47	3	38	1